

→ Full item /
Nothing

0-1 Knapsack Problem

Store N items, each item has a

- weight
- price

Thief is going to rob subset of items st total wt of items is
 $\leq$ capacity of thief's Bag.

Each item has 1 unit in stock

Find max amount the thief can make.

$$\underline{N=4}$$

	<u>Mall</u>			
				
wt	12	3	6	15
Price	<u>100</u>	<u>80</u>	40	90



$$\left\{ \begin{array}{l} 100 \\ \text{Box} \\ \hline (12) \end{array} + \begin{array}{l} 40 \\ \text{Bat} \\ (6) \end{array} = 140 \text{ ₹} \right.$$

18kg

$$\left[\begin{array}{l} \text{Ball + Duck (18kg)} \\ 90 + 80 = ₹170 \end{array} \right.$$

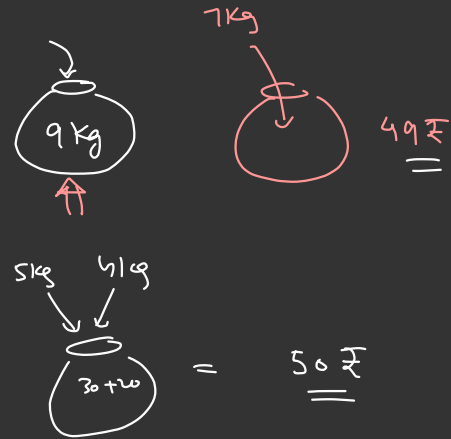
$$\checkmark \left[\begin{array}{l} \text{Bird + Box (15kg)} \\ 100 + 80 = \textcircled{180 \text{ ₹}} \end{array} \right.$$

↖

Greedy

- Pick the costliest item
- Pick item with more value / kg

N=3	wt	<u>7</u>	<u>5</u>	<u>4</u>
	Price	<u>49</u>	30	20
	unit price	<u>7 ₹</u>	6 ₹	5 ₹
			✓	



greedy fails!

↳ work if we are allowed to take a "fraction" of item

Brute force

(generating possible subsets)

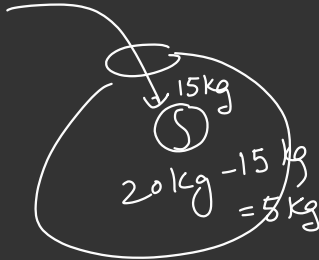
	0	1	2	3	
					$N=4$
wt	12	3	6	15	
Price	<u>100</u>	<u>80</u>	40	90	$C=20\text{kg}$

Pick or Not Pick
no of items

how much profit?

$$f(n, C) = \max \left\{ \begin{array}{l} \text{price}[n-1] + f(n-1, C - \text{wt}[n-1]) \\ f(n-1, C) \end{array} \right. \quad \text{if } C \geq \text{wt}[n-1]$$

N^{th} item $\downarrow dx$ $\downarrow$ no of items

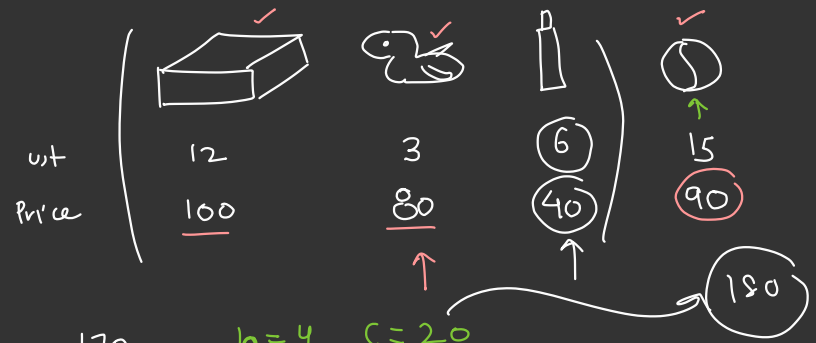


Recursive Case

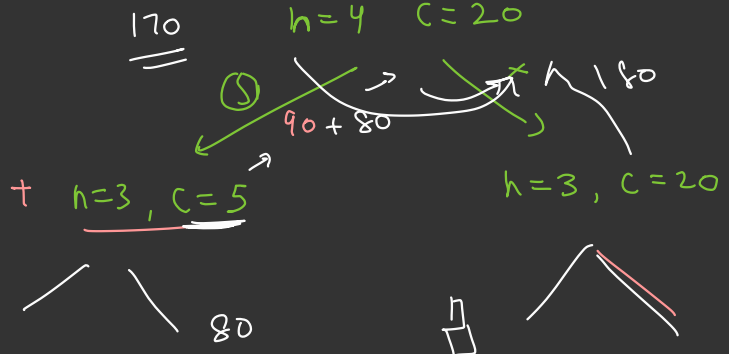
Base Case

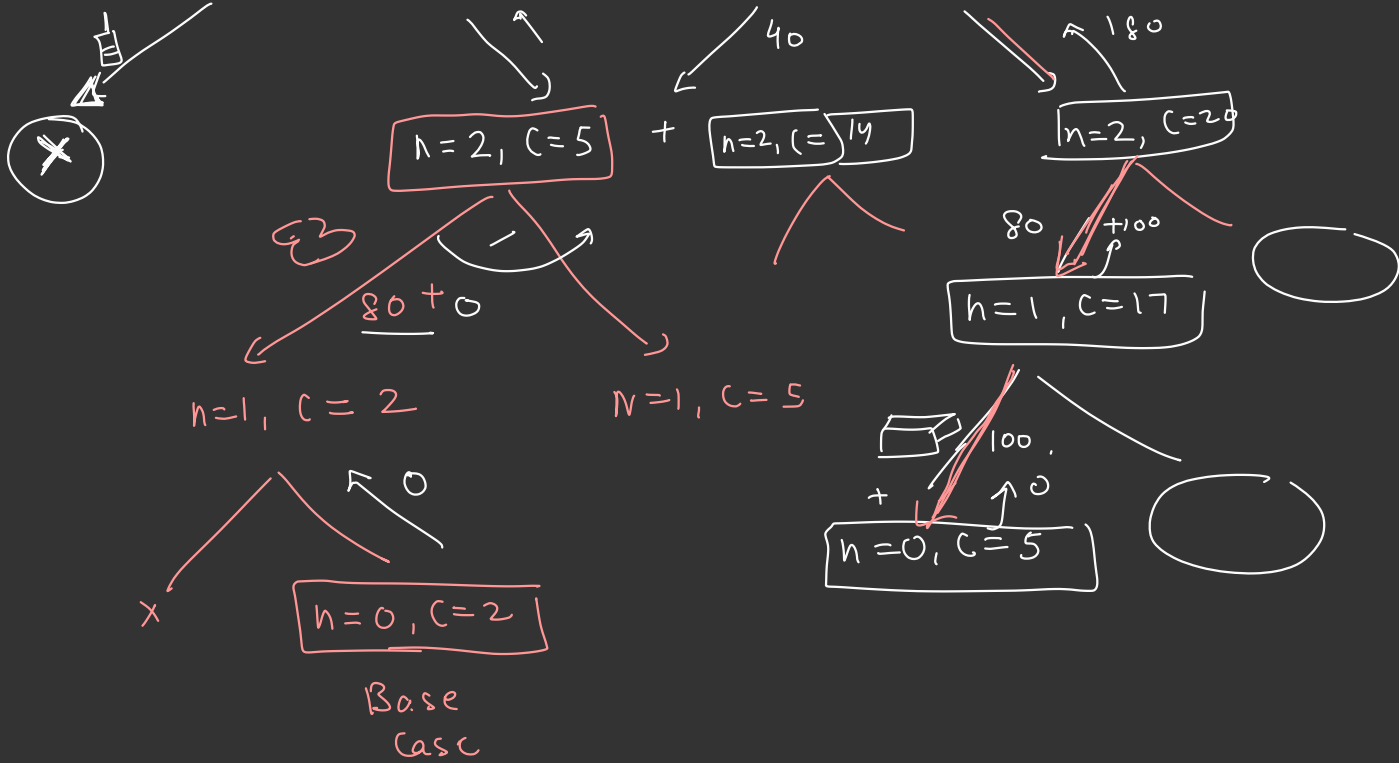
if (n == 0)
return n

if (c == 0)
return n



c = 20





Bottom Up

$dp[n][c]$ = Max Profit using upto n items and c capacity of Bag

$N=5$

$C=7$

$\begin{matrix} & \text{0} & \text{1} & \text{2} & \text{3} & \text{4} \\ \text{wts} & \text{3} & \text{6} & \text{5} & \text{2} & \text{4} \\ \text{prices} & \text{12} & \text{20} & \text{15} & \text{6} & \text{10} \end{matrix}$

Best 3kg, 4kg

$12 + 10 = 22$

$6 + \text{()}$
 $6 + \text{()}$ = 6
 = 12

$10 + dp[4][3]$

$dp \rightarrow n+1$ Rows, $C+1$ cols

$\Rightarrow$

		0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0	0
1	0	0	0	12	12	12	12	12	12
2	0	0	0	12	12	12	12	20	20
3	0	0	0	12	12	15	20	20	20
4	0	0	6	12	12	15	20	20	20
5	0	0	6	12	12	15	20	20	20

$n=0 \rightarrow 0$

if item $\text{knos wt} \leq c$

$dp[n][c]$

$= \max \left\{ \begin{array}{l} \text{price}[n-1] + dp[n-1][c - \text{wt}[n-1]] \\ dp[n-1][c] \end{array} \right.$

$dp[5][3] \rightarrow dp[4][3]$
 $dp[5][7] \rightarrow 10 + dp[4][3]$

$n=1$

$$dp[1][1] = \max \begin{cases} \text{ } \\ dp[0][1] \end{cases}$$

$$dp[1][2] = \max \begin{cases} \text{ } \\ dp[0][2] \end{cases}$$

$\uparrow \quad \uparrow$
 $n=1 \quad c=2$

$$dp[1][3] = \begin{cases} 12 + dp[0][3-3] = 12 \\ dp[0][3] = 0 \end{cases}$$

$$dp[1][4] = \begin{cases} 12 + dp[0][4] = 12 \\ dp[0][4] = 0 \end{cases}$$

$\vdots$

$n=2$

wt	3	6	5
p	12	20	15
		$\uparrow$	$\uparrow$

$$dp[2][1] = \begin{cases} \text{ } \\ dp[1][1] \end{cases}$$

$$dp[2][2] = \begin{cases} \text{ } \\ dp[1][2] \end{cases}$$

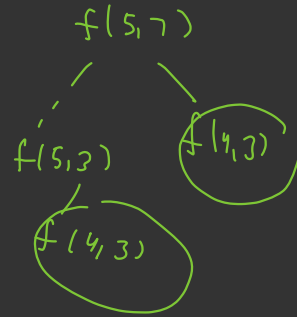
$$dp[2][6] = \begin{cases} 20 + dp[1][0] = 20 \\ dp[1][6] = 12 \end{cases}$$

$$dp[3][5] = \begin{cases} 15 + dp[2][0] = 15 \\ dp[2][5] = 12 \end{cases}$$

$$dp[2][3] = \begin{cases} \text{ } \\ dp[1][3] \end{cases}$$

$$dp[2][4] = \begin{cases} \text{ } \\ dp[1][4] \end{cases}$$

$$dp[2][5] = \begin{cases} \text{ } \\ dp[1][5] \end{cases}$$



overlapping
subproblems

$$dp[3][6] = \begin{cases} 15 + dp[2][1] & = 15 \\ dp[2][6] & = \end{cases}$$

$$dp[4][2] = \checkmark$$

$$TC \rightarrow O(NC)$$

$$SC \rightarrow O(NC)$$

"Space Optimised version" - just need to maintain prev row to calc
row

$$dp \rightarrow 2 \times (C+1)$$

$$\boxed{SC = O(C)} \leftrightarrow$$

→ no limit on qty available
Unbounded Knapsack

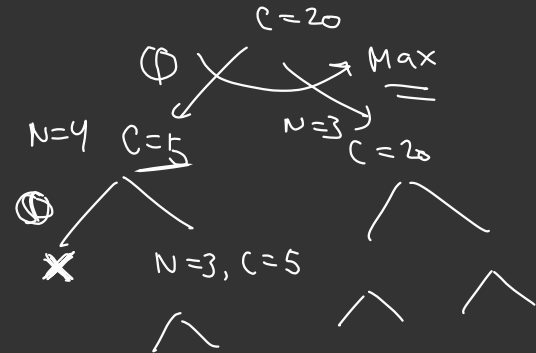
	<u>Mall</u>			
				
wt	10	3	6	15
Price	<u>100</u>	<u>80</u>	40	90
Stock	∞	∞	∞	∞

↑



6 Birds
 $6 \times 80 = 480 ₹$

greedy?



$f(n, c)$

Unbounded
Knapsack

$\left. \begin{array}{l} \text{inc} \\ \text{max} \\ \text{exc} \end{array} \right\}$

$\text{price}[n-1]$ +

$f(n, c - \text{wt}[n-1])$

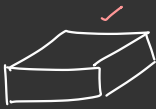
if $\text{wt} \leq \text{curr cap}$

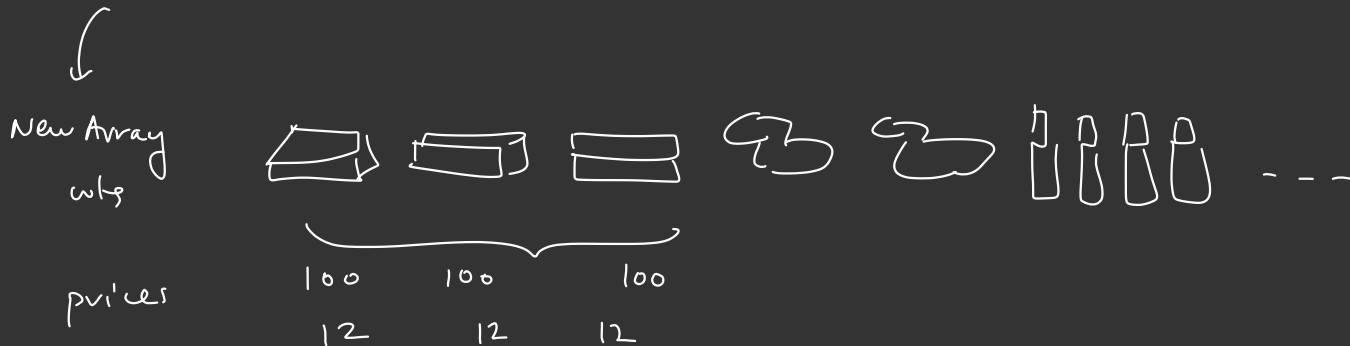
$f(n-1, c)$

$f(n-1, c - \text{wt}[n-1])$

↓
Earlier case

Scenario

	<u>Mall</u>			
				
wt	12	3	6	15
Price	<u>100</u>	<u>80</u>	40	90
Qty	3	2	5	1



0-1 KnapSack.

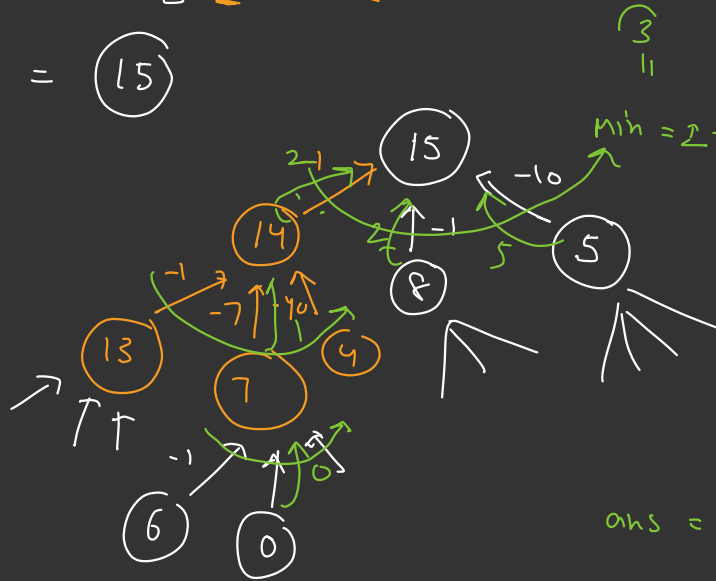
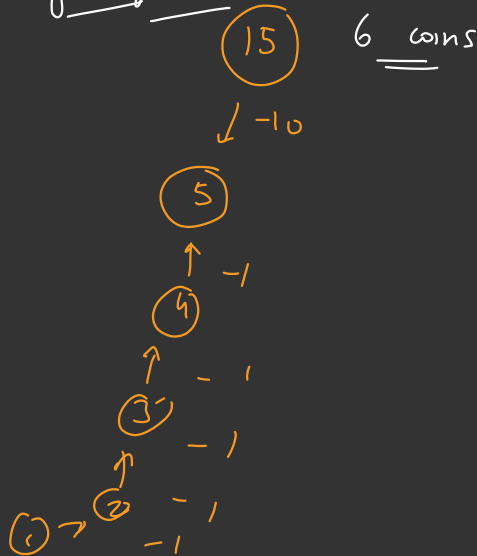
Coin Change Problem (1D DP)

Min coins needed to make the change

$$\text{coins} = [\underline{1}, 7, \underline{10}]$$

$$N = (15)$$

greedy fails



$$\min \left(\begin{matrix} x \\ \text{f}(N-1) \end{matrix}, \begin{matrix} y \\ \text{f}(N-7) \end{matrix}, \begin{matrix} z \\ \text{f}(N-10) \end{matrix} \right) + 1$$

$$1, 7, 7$$

if (N == 0)
return 0

$$\text{ans} = \min \left\{ \begin{matrix} \text{f}(N - \text{coins}(0)) \\ \text{f}(N - \text{coins}(1)) \\ \vdots \end{matrix} \right\} + 1$$

Top Down DP

int minCoins (int N, int[] coins) {

if (N == 0)
return 0

dp →

// Rec Case

ans = ∞

for (coin : coins) {

if (N - coin >= 0) {

subprob = minCoins (N - coin, coins)

ans = min (ans, subprob + 1)

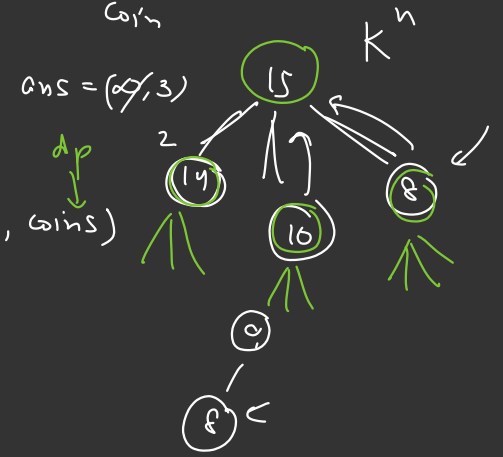
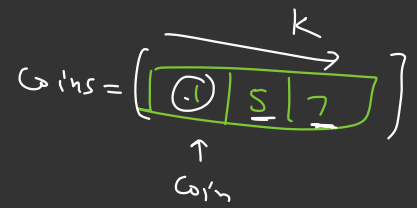
}

3

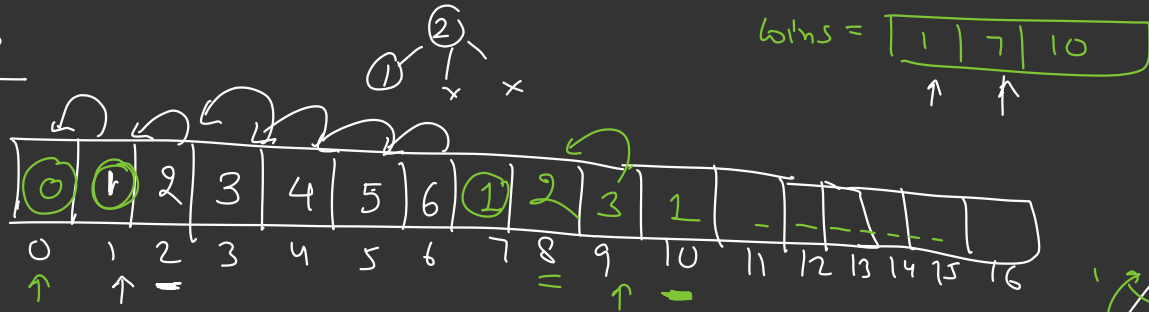
dp

return ans

4-5



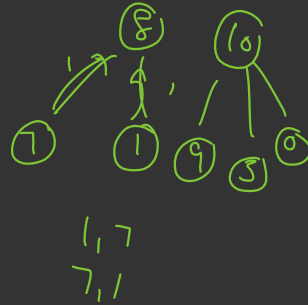
Bottom Up DP



$$dp[0] = 0$$

$dp[x] =$ min coins to make change for x

$$dp[7] = \min \left(\begin{array}{l} dp[1-1] \\ dp[1-7] \\ dp[1-10] \end{array} \right) + 1$$



for ($i=1$ _____ n) {

$$dp[i] = \min \left(\begin{array}{l} dp[i-1] \\ dp[i-7] \\ dp[i-10] \end{array} \right) + 1$$

→ loop

